Ryan C. Gelnett

Curriculum Vitæ

University at Albany, State University of New York – ryancgelnett.github.io – rgelnett@albany.edu

Areas of Research

Geometric Group Theory and Algebraic Topology

Education

University at Albany, State University of New York
Ph.D. Student in Mathematics
Expected May 2027

State University of New York at Oswego
Bachelor of Science in Applied Mathematics
Minor in Logic
Summa Cum Laude
May 2021

Papers and Preprints

On spaces of embeddings of circles in surfaces 2026
To appear in Michigan Mathematical Journal
arXiv:2601.12450

Configuration spaces of circles in the plane 2026
With Justin Curry and Matthew Zaremsky
Journal of Homotopy and Related Structures 21, no. 2 (2026): 277-305.
arXiv:2411.04800

Talks – Invited and Contributed

Invited

On spaces of embeddings of circles in surfaces

AMS Sectional Meeting–Syracuse University, (planned) Apr. 10-11, 2027

SUNY Oneonta, (planned) Feb. 19, 2027

UBDoG–University at Buffalo, Feb. 23, 2026

Contributed

On configuration spaces and their (co)homology groups

Algebra/Topology Seminar–SUNY Albany, Sept. 3, 2026

On spaces of embeddings of circles in surfaces

GSTGC–Rutgers University, Apr. 26, 2026

Configuration Spaces of Jordan Curves in Riemann Surfaces

BUGCAT–Binghamton University, Nov. 15, 2025

Point Selections from Jordan Domains in Riemann Surfaces

ATIA–SUNY Albany, Sept. 30, 2025

Configuration Spaces of Circles in the Plane

Young Topologist's Meeting–KTH Royal Institute of Technology, June 23-27, 2025

GTA Conference–Temple University, May. 31, 2025

ANYSRGMC–Syracuse University, Mar. 28-29, 2025

UNYSTS–Binghamton University, Mar. 2, 2025

NEAT–Providence College, Nov. 2-3, 2024

BUGCAT–Binghamton University, Oct. 27, 2024

Algebra/Topology Seminar–SUNY Albany, Oct. 10, 2024

Extended Grassfire Transform on Medial Axes of 2D Shapes

ATIA–SUNY Albany, Jan. 7, 2025

Algebraic Topology

SUNY Oswego, Spring 2021

Lagrangian Mechanics

SUNY Oswego, Spring 2021

Differential Topology

SUNY Oswego, 2019

Polyominoes

MAA Seaway, 2019

Conferences, Schools, Seminars and Workshops Attended

Algebra / Topology Seminar	SUNY Albany, 2021 – 2026
Configuration Spaces Summer School	University of Michigan, May 25-29, 2026
Topology Festival	Cornell University, Spring 2026
GSTGC	Rutgers University, Spring 2026
Homological aspects of groups and their classifying spaces	SwissMAP Research Station, May 4-9, 2025
Focus Program on Algebraic Topology In memory of Fred Cohen	Fields Institute (Virtually due to occupation), July 23-25, 2025
Young Topologist's Meeting	KTH Royal Institute of Technology, June 23-27, 2025
ANYSRGMC	Syracuse University, Spring 2025
GTA Conference	Temple University, Spring 2025
BUGCAT	Binghamton University, 2021 – 2025
UNYSTS	Binghamton University, Spring 2025
NEAT	Providence College, Fall 2024
AMS Eastern Sectional	SUNY Albany, Fall 2024

Groups, Actions, and Geometries — Tufts University, Aug. 7-11, 2023

MAA Seaway — 2018 – 2022

Relevant Work Experience

Teaching

Elementary Statistics (Instructor) — SUNY Albany, Fall 2026

Calculus 1 (Differential Calculus, Instructor) — Two sections, SUNY Albany, Fall 2026
Online, SUNY Albany, Summer 2024
SUNY Albany, Spring 2024

Calculus 3 (Multivariable Calculus, Instructor) — Online, SUNY Albany, Summer 2026
SUNY Albany, Fall 2025

Topics in Contemporary Mathematics (Instructor) — SUNY Albany, Spring 2026

Tutoring — SUNY Albany, Fall 2021 – Spring 2026
SUNY Oswego, Fall 2018 – Spring 2021

Educational Opportunities Program (EOP, Instructor) — SUNY Oswego, Summer 2025
SUNY Oswego, Summer 2024

Calculus 2 (Integral Calculus, Instructor) — SUNY Albany, Spring 2025
SUNY Albany, Fall 2024

Precalculus (Instructor) — SUNY Albany, Fall 2023
SUNY Albany, Fall 2022

Survey of Calculus (Grader/Teaching Assistant) — SUNY Albany, Spring 2023
SUNY Albany, Spring 2022
SUNY Albany, Fall 2021

Elementary Statistics (Grader/Teaching Assistant) — SUNY Albany, Spring 2023
SUNY Albany, Spring 2022
SUNY Albany, Fall 2021

EOP (Teaching Assistant) — SUNY Oswego, Summer 2019

Miscellaneous

Math Peer Leader — SUNY Oswego, Fall 2019

Genius Olympiad Assistant — SUNY Oswego, Summer 2019

Awards & Recognition

Louis R. DeRitter Award	SUNY Oswego, 2021
	SUNY Oswego, 2020
Presidents List (GPA of 3.80+)	Fall 2017 – Spring 2021

Other Activities

Moderated Talks	University of Michigan, Summer 2026
	MAA Seaway, Spring 2019
Math Club President	SUNY Oswego, 2019-2021
Math Club Secretary	SUNY Oswego, 2018-2019